

Theorem (Triangle Inequality). *Let V be an inner-product space over $\mathbb{R}$ or $\mathbb{C}$ with inner product $\langle \cdot, \cdot \rangle$ and norm $\|v\| := \sqrt{\langle v, v \rangle}$. For all $x, y \in V$ one has*

$$\|x + y\| \leq \|x\| + \|y\|.$$

Proof. Degenerate cases. If $x = 0$ then $\|x + y\| = \|y\| = \|0\| + \|y\|$. If $y = 0$ the same argument gives $\|x + 0\| = \|x\| = \|x\| + \|0\|$. Hence the inequality holds (with equality) when either vector is the zero vector. In the sequel we assume $x \neq 0$ and $y \neq 0$.

Squaring the inequality. Since norms are non-negative, $\|x + y\| \leq \|x\| + \|y\|$ is equivalent to

$$\|x + y\|^2 \leq (\|x\| + \|y\|)^2. \quad (1)$$

Using the definition of the norm we expand each side.

Left-hand side:

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2. \end{aligned} \quad (2)$$

(The equality $\langle x, y \rangle + \langle y, x \rangle = 2 \operatorname{Re} \langle x, y \rangle$ follows from conjugate-symmetry.)

Right-hand side:

$$(\|x\| + \|y\|)^2 = \|x\|^2 + 2\|x\| \|y\| + \|y\|^2. \quad (3)$$

Application of Cauchy–Schwarz. For all $u, v \in V$,

$$|\langle u, v \rangle| \leq \|u\| \|v\|,$$

with equality iff u and v are linearly dependent. Consequently,

$$\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle| \leq \|x\| \|y\|. \quad (4)$$

Deriving the squared inequality. Insert the bound (4) into (2):

$$\begin{aligned} \|x + y\|^2 &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 \quad \text{by (4)} \\ &= (\|x\| + \|y\|)^2 \quad \text{by (3)}. \end{aligned}$$

Thus (1) holds:

$$\|x + y\|^2 \leq (\|x\| + \|y\|)^2.$$

Taking square roots. Both sides of the previous inequality are non-negative, so taking the (non-negative) square root yields

$$\|x + y\| \leq \|x\| + \|y\|.$$

Equality case. Equality can occur only if every inequality used above is an equality. The algebraic expansions (2)–(3) are identities; the potentially strict step is the Cauchy–Schwarz estimate (4). Hence equality requires

$$|\langle x, y \rangle| = \|x\| \|y\| \quad \text{and} \quad \operatorname{Re} \langle x, y \rangle = |\langle x, y \rangle|.$$

The first condition is the equality case of Cauchy–Schwarz, i.e. x and y are linearly dependent: $y = \lambda x$ for some scalar λ (or one vector is zero). The second condition forces $\langle x, y \rangle$ to be a non-negative real number, which implies that λ is a non-negative real scalar.

Thus equality holds *iff*

- $x = 0$ or $y = 0$, or
- there exists $\lambda \geq 0$ such that $y = \lambda x$ (the vectors are positively collinear).

Therefore, for any vectors $x, y \in V$ the triangle inequality $\|x+y\| \leq \|x\| + \|y\|$ holds, with equality precisely in the cases described above. $\square$ $\square$